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Review

Current status of salt reduction in bread and bakery products – A review

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ABSTRACT

Cardiovascular disease, including hypertension, is the leading cause of preventable death worldwide. Sodium chloride (NaCl) associated with its excessive consumption is linked to hypertension while the global dietary NaCl intake has increased extensively. Thus, limitations on sodium (Na) consumption were recommended by the WHO and further international health agencies (<2 g Na/day).

Bread and other cereal products contribute about 30% to the daily intake of sodium in the western human diet. Although it may sound simple, salt reduction in foods is not as straight forward since salt impacts the processability during the production and the quality characteristics of the final bakery products. To achieve the final reduction goals long-term strategies and reformulation of recipes are required. Numerous different techniques have been proposed to reduce the sodium chloride content. Other approaches include salt replacers or enhancers. These are summarized with special attention given to recent developments over the last 5 years. One of the promising strategies to reduce salt has been the addition of sourdough to bakery goods. Sourdough can counteract some of the negative impacts salt reduction has on bread (e.g., flavour, shelf life) improving the overall quality.

The present review describes and assesses the impact of NaCl reduction on a range of bakery products, including dough characteristics, sensory properties and shelf-life. It further outlines the progress that has been made in salt reduction and indicates existing problems in the bakery sector.

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Abbreviations

ACE	Angiotensin I converting enzyme
AISBL	International Assocoiation of Plant Bakers
a_w	water activity
BU	Brabender Units
CASH	Consensus Action on Salt and Health
CHD	Coronary Heart Disease
CIQUAL	French Information Center on Food Quality
CVD	cardiovascular disease
CVUA Stuttgart	Chemisches und Veterinäruntersuchungsamt, Stuttgart
DoH	UK Department of Health
E′	extensibility
EC	European Commission
ENaC	epithelial Na+ channel
EU	European Union
FAO	Food and Agriculture Organisation
FSA	Food Standards Agency of the UK
FSAI	Food Safety Authority of Ireland
G′	storage modulus
GABA	γ−aminobutyrate
GDA	guideline daily amount
KCl	potassium chloride
LAB	lactic acid bacteria
Rmax	resistance to extension
TOHP	trials of hypertension prevention
TONE	randomized controlled trial of nonpharmacologic interventions in the elderly
UK	United Kingdom
US	United States (of America)
WHO	World Health Organisation

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1. Introduction

Increased consumption of dietary sodium is a leading cause of cardiovascular disease (CVD) and hypertension and has also been linked to an increased risk of stroke, stomach cancer, kidney disease and bone demineralization (Wardener and MacGregor, 2002). According to the American Heart Association CVD and stroke are the first and second-leading global causes of death which, together with their related diseases, account for 17.3 million deaths per year (Mozaffarian et al., 2015). High blood pressure has been noted as one of the major modifiable factors in the development of CVD. Several epidemiological, genetic, animal and population based intervention studies have established high sodium chloride intake as a factor in the increase in blood pressure with age in industrialised countries (He and MacGregor, 2007). From a public health perspective, this makes controlling hypertension related to sodium intake an important issue. Numerous national and international organisations have introduced recommendations and actions for lowering sodium chloride levels in foods. Among others the WHO and EU have recommended cooperation with the food industry to encourage the reduction in sodium content of products to their lowest feasible level. The recent National Adult Nutrition Survey conducted in the Republic of Ireland (Walton, 2011) revealed that between 65 and 70% of sodium chloride in the diet comes from processed foods, of which 22% originates from bread and bread products. The World Health Assembly has approved a set of nine voluntary targets for reducing behavioural and physiological risk factors for non-communicable diseases, including a target of reducing mean population sodium chloride intake by 30% to ≤5 g/day by 2025 (WHO, 2013). Those particularly vulnerable to high sodium chloride intakes are those over the age of fifty, people with high or elevated blood pressure and diabetics. In order to reduce salt intake effectively in the general population, EU salt reduction activities have concentrated on a limited number of food categories. Bread, classed worldwide as a staple food, is the only bakery product listed within these 12 priority categories.

The Food Standards Agency of the UK (FSA) and the Food Safety Authority of Ireland (FSAI) began their salt reduction programmes in 2003 when Irish bread contained average 1.2% salt (Lynch et al., 2009). Since then, the bread producing industry in Ireland and the UK has been actively trying to reduce sodium content in their products; however, its role in both the sensory and technological aspects of the bread product must be taken into consideration. The periodic monitoring over the last 12 years showed that the concentration of salt in processed food was reducing towards the set targets (FSAI, 2015).

Generally, many food additives contain sodium such as baking powder and soda which are used as leavening agents in various baked goods. Furthermore, sodium chloride is added to bakery products to improve taste, flavour and aroma (Cauvain, 2007a). Moreover, it plays a vital role in processing by improving textural properties and water binding capacity of the dough (Beck et al., 2012a; Belz et al., 2012a; Lynch et al., 2009). Sodium chloride also acts as a preservative against microbial spoilage by reducing the water activity (Hutton, 2002).

This review illustrates the current situation in salt reduction, problems arising in the bakery sector as a consequence of such reduction, and provides possible solutions (Fig. 1).

2. Sodium chloride and health

2.1. Sodium chloride

Sodium chloride (NaCl) is composed of sodium (Na^+) and chloride (Cl^-) ions joined by an ionic bond. Sodium chloride is essential for life and is better known as (table) salt. Throughout the following review, the terms ‘salt’ and ‘sodium chloride’ are not synonymous with the term ‘sodium’. For the conversion of sodium

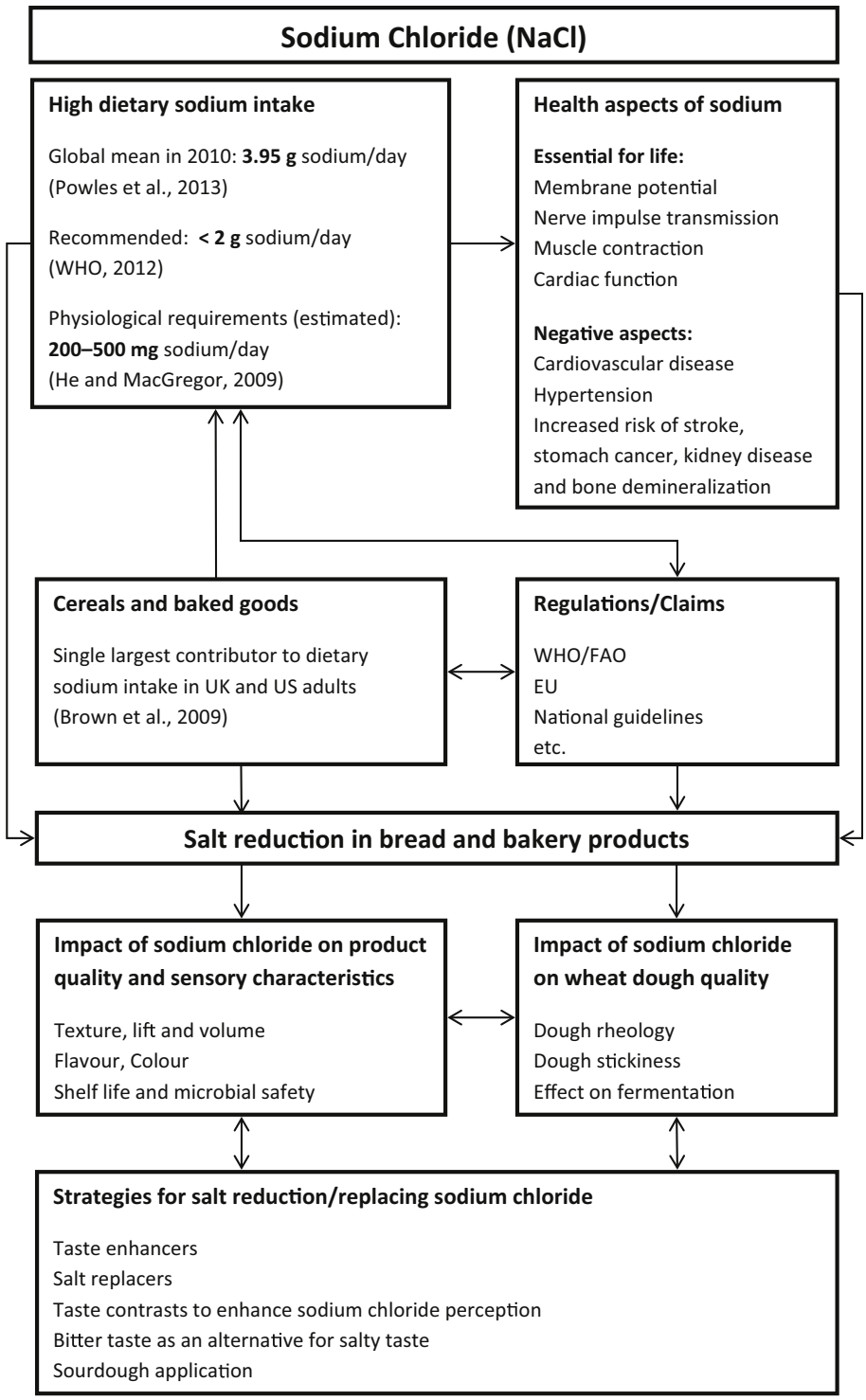

Fig. 1. Overview of salt and salt reduction in bread and bakery products.

into sodium chloride its weight must be multiplied by 2.54. Tight regulation of the body's sodium and chloride concentrations is an important biological process, with multiple mechanisms working to control it. Sodium chloride is required for survival, however health implications of excess sodium chloride intake represent an area of continued investigation among scientists, clinicians and public health experts.

Sodium and chloride, which are principal ions in extracellular fluid, which includes blood plasma, play a critical role in life-sustaining processes. They maintain membrane potential which is critical for nerve impulse transmission, muscle contraction and cardiac function. The absorption of Na^+ in the small intestine plays an important role in the absorption of chloride, amino acids, glucose and water. Similar mechanisms are involved in re-absorption of these nutrients after filtration from the blood by the kidneys.

Only a few studies have investigated the effects of sodium reduction on cardiovascular disease and on mortality, with mixed results. In general, the studies suggest a direct association, particularly those studies that used urinary sodium as a measure of sodium intake (Nagata et al., 2004). The TONE study (a randomized controlled trial of nonpharmacologic interventions in the elderly) indicated a trend toward reduced cardiovascular disease in participants assigned to the sodium reduction intervention (Whelton et al., 1998). A subsequent study found that participants initially without hypertension who were enrolled in the sodium interventions in the two previous trials of hypertension prevention (TOHP) had 25% reduction in cardiovascular events 10–15 years later compared with the control groups (Cook et al., 2007). Analyses from this TOHP follow-up study showed that the sodium-potassium ratio was associated with increased risk of cardiovascular disease in a dose-response relationship (Cook et al., 2009), providing complementary evidence for the adverse association between sodium chloride intake and cardiovascular disease.

2.2. Recommended and actual levels of sodium intake

Currently the WHO/FAO provides a guideline of <2 g of sodium intake per day (WHO, 2012). Most countries also have their own guidelines on sodium intake. In the US the food and nutrition board have approved dietary reference intakes of 1.5 g sodium per day with an upper limit of 2.3 g for those aged 18–50, 1.3 g for those aged 51–70 years and 1.2 g sodium per day for those over 70 years (Institute of Medicine (US), 2008). Similarly the UK suggests limiting sodium to 2.4 g per day (Scientific Advisory Committee on Nutrition, 2003), with this level being established as a guideline daily amount (GDA). Currently physiological requirements for salt (not well defined) are estimated to be as low as 0.5–1.2 g/day (200–500 mg Na) most of which will be excreted by the kidneys (He and MacGregor, 2009).

Several studies such as Brown et al. (2009) and Powles et al. (2013) give a comprehensive overview of the actual salt intake worldwide. In 2009 about 75% of sodium intake in European and Northern American countries was dominated by sodium added in manufactured foods, while cereals and baked goods were the single largest contributor to dietary sodium intake in UK and US adults (Brown et al., 2009). Around the world, with some exceptions in Africa and Asia, most adult populations have mean sodium intakes well in excess of 2.3 g/day (Brown et al., 2009). In contrast to the WHO advice, in 2010 the global mean sodium intake was 3.95 g/day (10.06 g/day of salt), nearly twice the WHO recommended limit (Powles et al., 2013).

Salt reduction initiatives have already resulted in 1.1 g of salt being removed from the Irish diet. Coronary Heart Disease (CHD) mortality rates in Ireland fell by over two-thirds in men and women between 1985 and 2006 (Brown and Jennings, 2014). About 28% of this decline was attributable to population blood pressure fall besides improvements in treatment uptake and risk factor improvements such as smoking, cholesterol and more physical activity. Blood pressure in turn is linked to sodium intake as discussed above. To achieve the UK recommendation of 6 g salt a day by 2015, in the early 2000s a voluntarily program was started where all members of the food industry were encouraged to agree to the salt targets and work towards them within an acceptable timeframe. Every 2 years the targets for salt reduction on over 80 food categories were reset and thus many food products are now 20–40% reduced in salt compared to 10 years ago (CASH, 2015). Due to the first achievements of the salt reduction programs, the UK Department of Health (DoH) has set new reduction targets for a total of 76 food groups, including breads and cakes, for 2017 (DoH, 2014). These new targets include an additional 10% reduction to a salt average of 0.9 g per 100 g in bread and rolls by 2017 compared to the 2012 target (1 g per 100 g). Furthermore, new reduction targets for salt in cakes (down from 0.5 g to 0.43 g per 100 g) and pastries (down from 0.5 g to 0.35 g per 100 g) were presented (DoH, 2014).

According to the Center for Science in the Public Interest (2014) reducing sodium consumption by half in the US would save an estimated 100,000 lives per year and would reduce medical care and other costs by roughly $1 trillion over the next 10 years. Even a modest gradual reduction of 1 g per day between 2010 and 2019 would be more cost-effective than using medications to lower blood pressure in all persons with hypertension (Bibbins-Domingo et al., 2010).

The effects of sodium and its levels are controversially discussed. Sodium reduction might well benefit those at the very high end of sodium intakes as can be seen from the substantial data available. According to Alderman and Cohen (2014) there must be a broad 'safe zone' of sodium intake (about 2.5–5 g/day). As with other essential nutrients, there is evidence of increased morbidity and mortality above and below a safe mid–range intake. It may be harmful if sodium reduction is pushed too far. Thus, it is also important to identify the dimensions of a 'safe zone' of sodium intake.

2.3. Regulations

Worldwide, several political institutions like the European Union have taken action to actively promote a reduction in salt levels in foods. In 2006 the European Parliament and the Council of the EU passed a regulation on "nutrition and health claims made on foods" (EC) No 1924/2006 (European Commission (EC), 2006). This document allows, amongst others, the use of nutrition claims regarding the content of sodium/sodium chloride in foods. Article 8 of regulation (EC) No 1924/2006 lists the following restrictions for claims on sodium/sodium chloride in its Annex:

Reduced (sodium/sodium chloride)

"A claim stating that the content in one or more nutrients has been reduced, and any claim likely to have the same meaning for the consumer, may only be made where the reduction in content is at least 30% compared to a similar product, except […] for sodium, or the equivalent value for salt, where a 25% difference shall be acceptable."

Low sodium/sodium chloride

"A claim that a food is low in sodium/sodium chloride, and any claim likely to have the same meaning for the consumer, may only be made where the product contains no more than 0.12 g of sodium, or the equivalent value for sodium chloride [0.3 g], per 100 g or per 100 ml."

Very low sodium/sodium chloride

"A claim that a food is very low in sodium/sodium chloride, and any claim likely to have the same meaning for the consumer, may only be made where the product contains no more than 0.04 g of sodium, or the equivalent value for sodium chloride [0.1 g], per 100 g or per 100 ml."

Sodium-free or sodium chloride-free

"A claim that a food is sodium-free or sodium chloride-free, and any claim likely to have the same meaning for the consumer, may only be made where the product contains no more than 0.005 g of

Table 1
Reported sodium chloride/sodium levels in bread and bakery products.

Country	Reference	Year	Bread type	Sodium level (mg/100 g	Equivalent as sodium chloride (g/100 g)
Australia	Dunford et al. (2011)	2011	All breads (mean)	435	1.1
Canada	Health Canada (2011)	2011	Bread (all types)	651–680	1.65–1.73
Canada	Health Canada (2011)	2011	Bagel (all varieties)	442–502	1.12–1.28
New Zealand	Dunford et al. (2011)	2011	All breads (mean)	435	1.1
US	Cauvain (2007b)		US pan bread		1.04
European Union					
France	CIQUAL (2012)	2012	Bread, country style	669	1.70
France	CIQUAL (2012)	2012	French baguette	681	1.73
France	CIQUAL (2012)	2012	Croissant	471–582	1.12–1.48
France	CIQUAL (2012)	2012	Puff pastry, sweet	382	0.97
Germany	CVUA Stuttgart (2015)	2014	Bread, bread rolls (all types)		1.0–2.9
Ireland	FSAI (2015)	2013	White bread	438	1.10
			Brown bread	428	1.09
Spain	Ballesteros (2010)	2009	Bread (all types)	642	1.63
UK	Brinsden et al. (2013)	2011	White & Brown		0.98 ± 0.13
UK	Cauvain (2007a)		UK fruit buns		0.72 ± 0.86
UK	Cauvain (2007a)		UK plain cake		0.39 ± 0.52
UK	Cauvain (2007a)		UK fruited cake		0.32 ± 0.42

sodium, or the equivalent value for sodium chloride [0.013 g], per 100 g."

Food companies favour a 'quiet' approach; while the use of salt replacers is growing, low salt claims are not. Low/no/reduced sodium claims are rarely used and appear on less than 4 percent of food and drink products launched globally (less than 1 percent in Europe and Asia) between 2010 and 2014 (Gray, 2014). This is an indication for a stealthy sodium reduction over time where changes are less noticeable and more palatable.

2.4. Consumption and current sodium chloride levels in bread and bakery goods

Bread belongs to those foods that form the basis of many civilizations' diets due to its nutritive value, its low price, and the simplicity of using its primary ingredient, the cereals, for culinary purposes. Bread is rich in complex carbohydrates (starch being the major component) and fibre, it has a high content of plant proteins, and contains hardly any fat. It is a good source of B vitamins and minerals such as phosphorus, potassium, and magnesium. All these components are particularly enriched in bread made from wholemeal flour, where the bran fraction is reintroduced into the milled white flour. Due to these nutritional properties, it comes as no surprise that nutrition experts define bread as an essential part of the food pyramid's base, as it should also constitute the base of the diet (in countries where bread is a staple source of carbohydrates).

In 2010, bakery product sales, including frozen bakery and desserts, reached a value of $145.9 billion in Western Europe (Agriculture and Agri-Food Canada, 2011). Average bread consumption in Europe is approximately 59 kg per person per year (AISBL, 2015). Bread consumption patterns differ widely between countries, with the highest reported being Turkey (app. 104 kg) and Bulgaria (app. 95 kg), while the lowest consumption is reported in the UK (app. 32 kg) (AISBL, 2015). Although bread is relatively low in sodium chloride, its high contribution to sodium chloride intake is due to the high quantities of this product consumed by the general population.

Globally there are wide variations in the levels of sodium chloride in bread and bakery products (Table 1). There is a difference of over 1 g of sodium chloride between the lowest country (UK) and the highest (Germany). Taking the UK as a case study, it is evident that a reduction in sodium chloride to an average of 0.98 g has not had any adverse effects on consumer acceptability and bread consumption as a whole (He et al., 2014).

Health Canada Guidance for the Food Industry suggests for pantry bread and rolls, bagels, croissants and flatbreads a reduction from currently 430 mg–330 mg sodium per 100 g by the end of 2016 (Bureau of Nutritional Sciences (2012)).

3. Techno-functional impact of sodium chloride on wheat dough quality

Sodium chloride makes wheat gluten more stable and less extensible, making it less sticky. It also affects the fermentation rate, reducing the rate of gas production by yeast. Inadequate amounts of sodium chloride result in loaves with open grain and poor texture (Hutton, 2002).

Cauvain (2007b) has summarized the main technological functions of sodium as:

- Impacting the development of gluten structures in the mixing of bread and other fermented products
- Inhibition of bakers' yeast in the fermentation of (bread) doughs resulting in a lower gas release
- Control of water activity in the baked product and thus a reduction of spoilage

The development of an extensible gluten network in the dough is a fundamental feature of the bread making process. This network contributes to gas cell formation and their subsequent retention during the fermentation and baking stages (Cauvain and Young, 2006). Hydration of the flour with water initiates the gluten network formation, and this is further developed by mixing which imparts energy into the dough. During mixing there are marked changes in the rheological properties of the dough mixture. It has been suggested that sodium chloride delays gluten hydration and thus increases mixing time of the dough, however there had been no direct evidence to confirm this hypothesis. In the case of puff pastry dough, the protein structure needs not be fully developed during mixing as the force of the subsequent lamination process is absorbed by the dough and thus the gluten network (Cauvain and Young, 2001). Since salt reduction weakens the gluten network it can cause problems during lamination. Thus, the dough layers, which need to stay intact, could break easily, as during baking the gluten in the dough layers serves to trap steam that results in the rising of puff pastry.

A recent study by McCann and Day (2013), examined dough microstructure in the early stages of dough development using

confocal laser scanning microscopy. Results indicated that the hydration of gluten protein and the development of the gluten matrix in the dough were delayed in the presence of NaCl, resulting in a gluten network with a more fibrous structure. Gluten protein consists of about 40% hydrophobic amino acids (mainly leucine and isoleucine) and a low proportion of charged amino acids at the protein surface. In the presence of NaCl some of the charges on these exterior amino acids may be shielded, therefore slowing down protein hydration. This reduction of protein surface charge reduces the electrostatic repulsion between the protein molecules allowing for them to interact more closely through strong hydrophobic interactions which increases protein intermolecular β-sheet structure (Wellner et al., 2005). The overall effect of increasing mixing time is to increase loaf height and improvement of overall bread quality. A reduction of sodium chloride levels in dough leads to a weakening of the gluten network, increasing dough stickiness. During the industrial manufacture of bread this can stop processing lines leading to down time and wastage (Wehrle et al., 1997).

The impact of salt is most readily observed in bread and fermented goods such as rolls and buns but also in laminated products such as croissant and puff pastry (Cauvain, 2007a). Where the gluten formation is limited or absent in cakes, biscuits, cookies and many pastry products there appears to be no significant effect of salt (Cauvain, 2007a). Therefore, these products are rarely investigated regarding salt reduction with many studies focused on the impact of salt reduction in bread (Belz et al., 2012a; Lynch et al., 2009; Noort et al., 2010).

3.1. Effect on dough rheology

The fundamental mechanical properties of wheat flour dough are important because they affect the dough handling behaviour during processing and also influence the interactions among dough components. To understand what is happening during dough formation rheological analysis are applied. Wheat dough is a diverse and complex material that exhibits viscoelastic properties. Gluten displays viscoelastic behaviour in which the gliadin fraction represents viscous behaviour and the glutenin fraction represents elastic behaviour (Edwards et al., 2001). The incorporation of sodium chloride into the wheat dough influences interactions between the gluten strands due to its ionic nature. In addition, sodium chloride is known to cause changes in water absorption which affects the formation of the viscoelastic network of dough.

Empirical methods apply simple elongation, shear or compression and are rapid, low cost, and simple to perform methods of measuring rheological properties. Farinograph measurements showed that for dough consistency at 500 Brabender Units (BU), NaCl addition produces a reduction in water absorption and an increase in dough development time (Hlynka, 1962). This is in agreement with subsequent studies carried out by Salovaara (1982) who compared salted wheat doughs using the farinograph and found that dough with 1–2% sodium chloride added had an increase in peak time (i.e. increase in dough strength), while water adsorption decreased (i.e. a decrease in water binding).

Extensibility is an important characteristic in wheat dough, particularly in the moulding step (e.g. bread) or lamination step (e.g. puff pastry). It is one of the characteristics used to relate the properties of a dough piece to its probable performance in processing and to possibly predict the end product quality. Lynch et al. (2009) used an extensograph to define extensibility and resistance to extension of doughs containing 0, 0.3, 0.6 and 1.2% sodium chloride. It was found that doughs that contained 0.3–1.2% sodium chloride did not differ significantly in their resistance to extension (R_{max}) and extensibility (E'). However, dough containing 1.2% sodium chloride significantly differed from dough containing 0% sodium chloride. These results would indicate that only omission of sodium chloride entirely would significantly affect dough extensibility.

Numerous studies have applied fundamental rheological measurements to investigate the effect of NaCl on dough rheology, however the results have been contradictory. Some studies found a decrease in the storage modulus G' with an increase in NaCl addition (Angioloni and Dalla Rosa, 2005; Lynch et al., 2009; McCann and Day, 2013). In contrast to this, a study by Beck et al. (2012b) reported an increase in G' when NaCl was added to the dough. This study showed that a reduced level of NaCl in dough changed the gluten protein network structure from elongated protein strands to less connected protein particles.

3.2. Effect on dough stickiness

Stickiness is a combination of adhesion, the interaction between a material and a surface, and cohesion, the interactions within the material (Wang et al., 1996). In industry dough stickiness is a major problem as sticky dough can lead to process disruption and product loss. Agraria (2009) gives an overview of the determination of stickiness -mainly expressed as work of adhesion-of dough including methods, parameters and the corresponding references.

Several studies showed that a reduction of sodium chloride lead to a significant increase of stickiness of wheat dough (Diler et al., 2016; Silow et al., in press). Moreover, the adhesive work decreased significantly as soon as a small amount of salt (0.35% d.b.) was added (Diler et al., 2016). Contrary, Beck et al. (2012b) reported a decreased dough stickiness with decreasing NaCl concentration (40–0 g/kg flour) in bread dough. Regarding the rheological properties of wheat dough including stickiness a salt reduction up to 25% in pizza (Diler et al., 2016) or even 30% in puff pastry (Silow et al., in press) was possible.

3.3. Effect on fermentation

A decrease in the level of salt added leads to an increase in gas production by the yeast. This is due to the decrease in the osmotic pressure and the electrochemical potential of sodium and chloride ions on the membrane of the yeast (Matz, 1992). The resulting dough may become 'over proved' in a standard proof time in the bakery why proofing time has to be adapted, whenever the salt level is changed (Lynch et al., 2009) observed a significant increase in the maximum dough height as the salt level decreased. At the same time an increase in the total volume of gas (CO_2) released was observed using a rheofermentometer. This indicates a weakening of the gluten network as the NaCl level was reduced. During proofing a continuously stretching of the dough gas cells in a biaxial way causes further development and hardening of the gluten network while the specific volume increases.

4. Impact of sodium chloride on product quality and sensory characteristics

4.1. Texture, lift and volume

Sodium is responsible for the formation of an even crumb (Matz, 1992). Lynch et al. (2009) reported much larger pores in bread without added salt than those containing 0.3–1.2% salt. In the course of proof and baking the growth of gas bubbles determines the expansion of the dough and ultimately volume and texture of the baked product (He and Hoseney, 1991). The limit of expansion of these bubbles is related directly to their stability, due to coalescence and the eventual loss of gas when the bubbles fail. The rheological properties of the bubble walls will therefore be

important in maintaining stability against premature failure during baking, and also in relation to gas cell stabilization and gas retention during proofing, and thus to the final structure and volume of the baked product (Dobraszczyk et al., 2001). The specific volume (volume to weight ratio) is directly linked to the volume of a baked good and is commonly used in the assessment of bread quality. Different trends can be seen for the specific volume of bread when salt levels were reduced. On the one hand, specific bread volume slightly decreased when no salt was added (Czuchajowska et al., 1989). On the other hand, Lynch et al. (2009) and Okano and Mizutani (1995) reported no significant difference but a trend towards increase in the specific volume of bread at reduced salt levels. For puff pastry with full and reduced fat content Silow et al. (in press) reported an increase of the maximal lift and the specific volume for increasing NaCl levels.

Major physical changes occur in all baked goods during baking. At temperatures above 100 °C steam and CO_2, if present, expand rapidly and cause high pressure on the surrounding cell walls. In these walls made of dough the gluten will trap the gases and retain them to a certain extent. Furthermore, starch starts to gelatinise, protein starts to denature (Cauvain and Young, 2001) and Maillard Reaction will yield browning products (see also 4.3). Finally, the dough layers form little cracks, the gluten cannot retain the gases anymore and they will diffuse or escape (Cauvain and Young, 2001). Salt reduction influences the texture of baked goods, in so far as salt impacts the gluten strength and if present the yeast activity.

4.2. Flavour

Sodium is responsible for overall aroma and flavour impression and can enhance sweetness (Ugawa et al., 1992) as well as mask bitterness (Breslin and Beauchamp, 1995). Only free Na^+ ions in solution which are available for sodium chloride taste perception on the tongue can elicit a sodium chloride taste. At the same time the contribution of the Cl^- ion on taste is not entirely clear yet. Due to extensive interaction between sodium chloride and wheat proteins during dough formation, a certain amount of sodium is bound due to ionic interactions with the negatively charged amino acids. As demonstrated by Pflaum et al. (2013a), neither Na^+ nor Cl^- ions are irreversibly bound in bread and it is only a matter of time until 100% of sodium ions are released by chewing. The velocity of sodium release during the first seconds of mastication seems to be rather important for the perceived saltiness. Among others, this release velocity depends on the crump structure (see also 5.3).

Bread without sodium chloride has been reported to be perceived as tasteless and have yeasty and sourdough like flavours (Lynch et al., 2009; Miller and Hoseney, 2008). The study by Pflaum et al. (2013a) revealed that an increase in sodium chloride level lead to a decrease in floury/watery, yeasty and musty flavours. When salt levels were reduced from 1.3% to 1% a difference in sodium chloride level could be detected, which were not detected at higher concentrations. This would indicate that 1% salt seems to be a critical concentration for sodium chloride detection.

4.3. Colour

Sodium chloride influences Maillard reactions which occur during baking. These are reactions between proteins or amino acids and reducing sugars resulting in the formation of compounds like melanoidins which help to form the crust colour of the bread and other baked products (Belitz et al., 2008). Various factors are influencing the formation rate and types of compounds in the final product, such as the pH, the protein/amino acids ratio, the water and sugar level, temperature and cooking time. It has been shown that sodium chloride has a plasticising effect during the heating of cereal products. This improves mobility of the reactants, enhancing the Maillard reactions, producing a darker coloured crust (Moreau et al., 2009). On the other hand, in absence of salt baked bread has a lighter coloured crust (Czuchajowska et al., 1989). Due to higher yeast activity more sugar is metabolised and not available for the formation of browning products (Skobranek, 1998). Thus, in non–fermented baked goods a reduced salt level should have no impact on the colour.

4.4. Shelf life and microbial safety

Sodium chloride prevents rapid spoilage, thus extends product shelf–life, and creates an inhospitable environment for pathogens. Sodium chloride is an effective preservative because it reduces the water activity (a_w) of foods. Increased osmotic pressure causes cells to lose water into the environment which in turn inhibits cell growth. In bread, sodium chloride helps control the growth of moulds and bacteria such as *Bacillius* species, thus extending shelf life (Betts et al., 2007). Challenge tests carried out with fungi commonly found in the bakery environment (such as *Penicillium expansium, Fusarium culmorum* and *Aspergillus niger*) exhibited a reduced shelf life of bread by 1–2 days when sodium chloride levels were decreased (Belz et al., 2012a). The use of lower sodium chloride levels must be compensated by the use of alternative anti–microbial strategies due to an increase in water activity. Generally, with a_w values between 0.96 and 0.98, bread is known as a high moisture product (Smith et al., 2004). In bakery products which are consumed within the baking day, like croissants or puff pastry, the higher water activity and spoilage issue can be mainly neglected. Apart from possibly moist fillings, a_w of these products is quite low even when salt reduced.

5. Strategies for salt reduction/replacing sodium chloride

Consumers are concerned more than ever about sodium content in their foods, but refuse to compromise on taste – and, of course, they also demand a clean label. Therefore industry aims to design ingredients for salt reduction which address these needs and keep the consumer–craved salty, hearty flavour.

Numerous different strategies have been proposed to reduce the sodium chloride content of foods. Methods to solve the technological, sensory and shelf–life problems associated with sodium chloride removal have been developed. Technologically, a reduction in salt to 0.6–0.3% would be feasible without a significant deterioration in rheological properties or performance during manufacture (Lynch et al., 2009). However, its effect on the organoleptic properties of bread is still a critical factor in consumer acceptance. With regards to this many strategies have been developed to improve taste quality.

Ideally the replacement of sodium is realised with compounds that evoke a similar pure saltiness when consumed. Such a replacement appears to be unlikely due the high specificity of sodium for the epithelial Na^+ channel (ENaC). Until now the whole mechanism of salt taste transduction has not been elucidated. Nevertheless, the ENaC seems to be very important for this mechanism, but cannot fully explain salt taste.

Partial replacement with inorganic salts such as potassium chloride (KCl) can work well up to 20–30% replacement, after which point metallic and bitter aftertastes are detected (Braschi et al., 2009).

Gradual reduction in sodium levels by stealth has been employed with some success which is reported by the Food and Drink Federation and British Retail Consortium (Wilson et al., 2012). The strategy ("small step reduction") aims to slowly and gradually reduce the salt content in recipes without the consumer's

notice. Girgis et al. (2003) found evidence to support this strategy by conducting an intervention study (110 participants). In that study, it was possible to reduce the salt content in bread by 25% and maintain consumer's acceptance. However this approach is limited as it takes a lot of time and can only be applied up to a critical point after which the product becomes unacceptable to consumers. Accordingly, the strategy also includes the use of spices, salt replacers and taste enhancers (Wilson et al., 2012). Due to these limitations the search for alternative strategies is still ongoing. More promising and recent approaches are the use of sourdough and taste contrast which could provide more effective solutions to salt reduction.

5.1. Taste enhancers

To overcome the blandness of low–salt bread, taste enhancers can be used. These include bread flavour preparations which can be purchased commercially, but also the addition of spices, malt or other flours like barley have been suggested to improve the flavour of low–salt breads (Rodbotten et al., 2015).

Specific salt enhancers are substances that do not have a salty taste in themselves, but enhance a sodium salty taste when used in combination with sodium chloride. A range of ingredients are reported to act as sodium chloride enhancers such as amino acids, monosodium glutamate, lactates, yeast products, soy based ingredients and other flavourings (Nakagawa et al., 2014). However, scientific data for bread are scarce. Taste enhancers activate taste receptors in the mouth and throat, which helps compensate for the salt reduction and enhance flavour. However, some of these salt enhancers may have consumer acceptability issues as they need to be stated on the ingredients list. Furthermore, they often cause changes in taste, aroma and also browning of the products.

5.2. Salt replacer (KCl)

The most commonly used salt replacer is potassium chloride, although other alternatives such as calcium chloride, magnesium chloride or magnesium sulphate have also been proposed (Kaur et al., 2011). Considering the metallic–bitter aftertaste of KCl, various studies have evaluated the levels of which salt can be replaced by KCl in bread recipes and are briefly summarized by Quilez and Salas-Salvado (2012). The group of Mueller et al. (2016) demonstrated for pizza crust that it was possible to replace 30% of sodium chloride by KCl without a noticeable loss of salty taste.

Shelf life tests on wheat bread revealed similar antifungal activity, when salt was replaced up to 30% with a mixture of various salt replacers (Samapundo et al., 2010).

5.3. Enhancement of sodium chloride perception by taste contrast

A study by Meiselman and Halpern (1973) showed that the delivery of continuously alternating concentrations of aqueous sodium chloride solutions enhance saltiness intensity compared to model solutions delivered in a non–alternating fashion. Thus, contrasting sodium chloride intensities can be used to enhance sodium chloride perception while still reducing the tastant. There are two main methods that have been developed to inhomogenously distribute sodium chloride in bread.

Noort et al. (2010) spatially separated different sodium chloride concentrations using horizontally laminated bread dough. This method does have a disadvantage in that production lines would have to be altered for manufacture on an industrial scale, which would be an added cost. Furthermore, the amount of yeast in each portion of dough needs to be adjusted to obtain constant fermentation speed between the doughs, independent of actual fermentation speed within a dough. Noort et al. (2012) used encapsulated *salts* to enhance sodium chloride perception by taste contrast. The average saltiness of the breads prepared with 1% large sodium chloride encapsulates (1000–2000 μm) were found to be equal or higher compared to those prepared with 2% normal sodium chloride. The disadvantage of using encapsulated *salts* is that there is no NaCl available to interact with the flour during mixing, therefore having a negative effect on dough development and also on the yeast fermentation rate, water activity and on the final bread texture and volume.

The use of coarse–grained sodium chloride showed the same result (Konitzer et al., 2013) causing increased contrast of sodium concentration which is known to trigger salt perception (Busch et al., 2009). The use of coarse–grained salt in bread making led to an inhomogeneous spatial distribution of sodium in the crumb. This resulted in accelerated sodium release and led to enhanced sodium chloride taste, allowing a sodium reduction in bread by 25% while maintaining taste quality. This sodium chloride taste enhancement can be explained by the increasing contrast in sodium concentration, which is known to determine sodium chloride taste perception. The same strategy also allowed successfully reduction of the salt content in pizza crust up to 25% (Mueller et al., 2016).

Pflaum et al. (2013b) reported the direct impact of the crumb texture on the saltiness perception. The bread crumb was impacted by different proofer conditions and the various crumb cell structures, particularly the more coarse-pored the bread crumb, resulted in enhanced saltiness perception. Considering that the use of sourdough can also change the crumb texture suggests that use of sourdough technology could positively influence low–salt bread.

5.4. Is bitter the new salty?

According to the recent Mintel's report (Mogelonsky, 2015) there is a growing consumer interest in bitter flavours. Sour and bitter flavours are added by industry to products reduced in sodium and sugar. In 2014 the American Heart Association reported that snack sales dropped by 40–50% when manufacturers reduced sodium. Thus, food and beverage companies added sweeteners to compensate for this sodium reduction. That was, however, rather counterproductive, since consumers generally pay more attention to their diet and search for healthier options.

Particularly in salty snacks "bitter" is the flavour profile which is becoming increasingly popular in US foodservice menu items. "Salty" is a flavour that is dominant in snack markets and subcategories whereas sweet flavour profiles are not as common and usually fall into the confectionery or bakery categories. Since 2013, only 5% "sweet" snack products have been launched, such as cereal bars and popcorn (Mogelonsky, 2015). Moreover, compared to 2014 in 2015 34% of US adults were limiting their intake of sweet snacks, and 33% were looking for healthier foods (Mogelonsky, 2015).

This is a big opportunity for bitter snacks to grow their share as consumers become more familiar with bitter–tasting food. An increase in menu items that are described as "bitter" can be observed in the US foodservice category, including cauliflower, Brussels sprouts and a range of bitter greens such as collards, kale, chicory, and mustard greens. Snacks with a bitter flavour profile may be less common, but they are gaining popularity, e.g. kale and root vegetable–based chips. The beverage industry is also tapping into bitter flavours. Bitter flavoured beverages, including bitter lemon, herbal cocktail bitters, matcha green tea, and beer formulations are becoming more popular with restaurant diners (Mogelonsky, 2015).

"Bitter" as a strong flavour note, may be one way of reducing the use of sodium in snacks and various other foods. In the past many

foodservice trends found their way into retail, bitter–flavoured products may do likewise.

5.5. Sourdough application

Sourdough is a fermented mixture of flour and water and has traditionally been used as a leavening agent in bread for thousands of years. In more recent times sourdough has been reintroduced as a functional ingredient to improve bread quality and to replace additives. Sourdough influences all aspects of bread quality. Technological effects of sourdough on the flavour, texture, shelf life (delay of staling, spoilage prevention), and nutritional quality of bread are well established and summarized in several recent reviews (Axel et al., n.d.; Galle and Arendt, 2014; Gänzle, 2014; Gobbetti et al., 2014). Conclusively, sourdough as an ingredient in bakery products favours the demand for clean label, natural products including a reduced use of additives. This is also more widely accepted by today's health conscious consumer.

The benefits of sourdough mentioned earlier may also be positively applied in salt reduced products to create functional foods. Belz et al. (2012b) suggested the highly promising potential of sourdough use to compensate the effect of salt reduction on flavour and the consequent influence on further sensory characteristics, like crumb texture.

The same research group demonstrated that the addition of sourdough, fermented with the antifungal *Lb. amylovorus* DSM19280, to salt–reduced bread prolonged the shelf life when compared to the control (Belz et al., 2012a).

Bread containing lactic acid bacteria (LAB)–fermented wheat germ tasted more salty when compared to control bread (Rizzello et al., 2010). The higher saltiness was assumed to be a combined effect of acidification and proteolysis. Due to the addition of rye malt sourdough fermented with glutamate–accumulating *Lb. reuteri* strains, it was possible to reduce the salt content in bread from 1.5 to 1% (flour base), keeping taste and other quality features unchanged (Zhao et al., 2015). A trained panel judged all sourdough breads to be also higher in umami and sour taste; this compensated overall for lower saltiness in NaCl–reduced bread, whereas the glutamate concentration met approximately the taste threshold for glutamate in bread; 0.03% monosodium glutamate (w/w) (Zhao et al., 2015).

Lactobacilli can further convert glutamate to γ–aminobutyrate (GABA) during fermentation by glutamate decarboxylase. GABA has several well–characterized bioactive functions, such as neurotransmission, diuretic effects, tranquilizer effects and induction of hypotension (Diana et al., 2014a; Li and Cao, 2010). The latter is the most important characteristic and has been studied intensively in human intervention trials (Hayakawa et al., 2002). During sourdough fermentation LAB may also synthesize Angiotensin I converting enzyme (ACE)–inhibitory peptides and antioxidant peptides (Coda et al., 2012; Hu et al., 2011; Rizzello et al., 2008). ACE–inhibitory peptides have shown to hinder the vasoconstrictory effects of Angiotensin II and to enhance the vasodilatory effects of Bradykinin which both result in a lowering of blood–pressure (De Leo et al., 2009). The benefits of GABA and ACE–inhibitory peptides were recently used by Diana et al. (2014b) and Peñas et al. (2015) to develop a sourdough bread targeted towards blood–pressure reduction. A liquid sourdough fermentation (whole wheat flour (16.5%), soya flour (2%), protease (1.5%); dough yield 500) was carried out using *Lb. brevis* CECT 8183 and additional protease. The thus produced low–sodium wheat–sourdough bread with potassium citrate (0.13%) contained 7 times more GABA at 24 mg/100 g and 3 fold higher peptide content (<3 kDa) at 7.5 mg/g when compared to the control (Peñas et al., 2015). The GABA concentration in commercial sourdough breads ranges from 2 to 9 mg/100 g (Diana et al., 2014b). A daily intake of 10–12 mg GABA for 12 weeks significantly reduced the systolic blood pressure ($p < 0.01$) in an intervention study with hypertensive human patients who consumed 100 mL of LAB–fermented milk (Inoue et al., 2003). Since bread is a staple food, the consumption of 100 g GABA–rich bread easily reaches effective levels. However, clinical studies are needed to prove the efficiency of the functional sourdough bread to decrease blood pressure in hypertensive patients. To increase the concentration of GABA in sourdoughs, studies have aimed to ferment protein rich flours from pseudocereals (quinoa, amaranth, buckwheat) and chickpea (Coda et al., 2010) or buckwheat sprouts (Nakamura et al., 2013).

In conclusion, sourdough enhances the salty perception as well as imparting additional flavour compounds. Due to the production of bioactive compounds like GABA, ACE–inhibitory peptides or antifungal substances, sourdough is a useful functional ingredient for salt-reduced breads. Moreover, the use of sourdough is not limited solely to bread alone. It may be incorporated into products other than bread. One possibility is the incorporation of sourdough in salt reduced puff pastry or croissants to improve their flavour, texture and therefore their palatability.

6. Conclusion

According to the WHO, reducing dietary sodium chloride is one of the most effective mechanisms to improve population health. Bread is a major contributor to sodium intake which means that ingredient reformulation and process adaptations are necessary to achieve this. Although some progress has been made further reductions are necessary. However this is difficult due to the important role of sodium chloride in bread making. To reduce sodium level in baked goods, it is first important to gain a fundamental understanding of its roles from both techno–functional and sensory aspects. Only after establishing this solutions can be developed to obtain comparable qualities in the product. Nevertheless, industry undertakes efforts to reduce the salt content in their products (Ahuja et al., 2015; Wilson et al., 2012). At least, it is also the responsibility of the consumer to more openly accept salt reduced products and avoid the general action of adding more salt to a product before consuming it (Zandstra et al., 2015).

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